UM
HACKATHON
2026

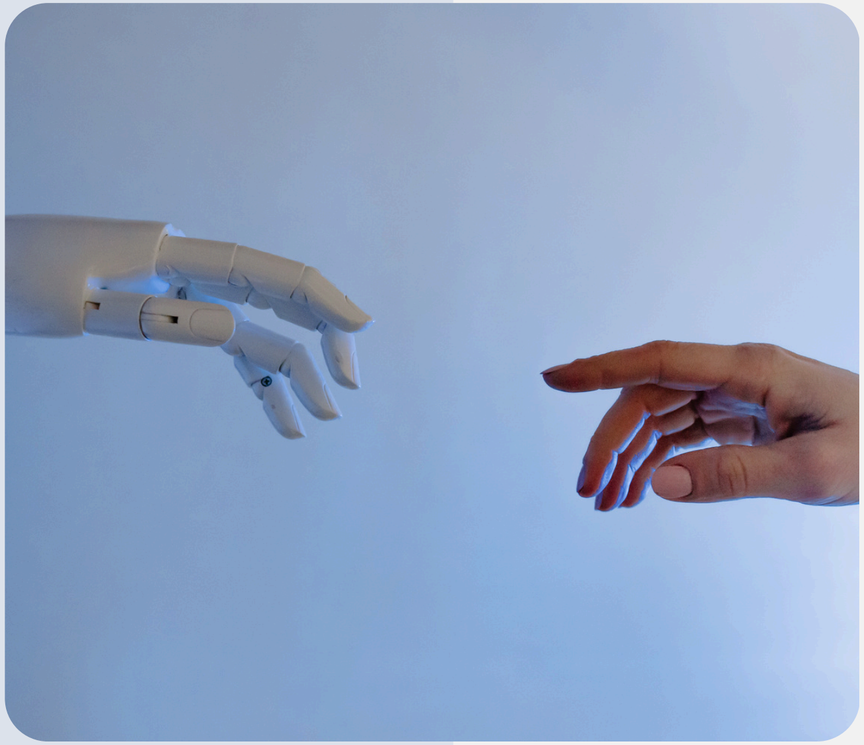
HOME PROBLEM BACKGROUND SOLUTION

Finite Automata

AI Fridge

GROUP MEMBERS:
IRFAN FATHI BIN HAMDAN
MUHAMMAD SYAMEER HARITH BIN MOHD YUSOF
MUHAMMAD AFIQ BIN MOHD FAUZI
MUHAMMAD FAHIM BIN JAAFAR
MUHAMMAD FARIS DINIE BIN NAZRUL ZAMRI

Problem Statement



The Food Waste Problem in Brief

Homes throw away 30–40% of their food because people lose track of what they have and when it goes bad.

Why current tools don't work:

- **Manual Operation:** Users currently have to manually record and update their fridge inventory, which is time-consuming
- **Lack of Visibility:** People simply forget what groceries they have, fail to track their inventory, and miss expiration dates.

Creating a single, automated system to fix this problem directly supports **Sustainable Development Goal 12,** helping the world cut food waste and use resources responsibly.

Background

Most food waste happens because people simply **forget** what they already own. Existing inventory apps require **manual entry**, which users abandon after days. Receipts, fridge contents and meal plans are not connected, forcing users to juggle multiple tools or rely on memory. This leads to duplicate purchases, spoiled food and wasted money.

Proposed Solution & Main goal

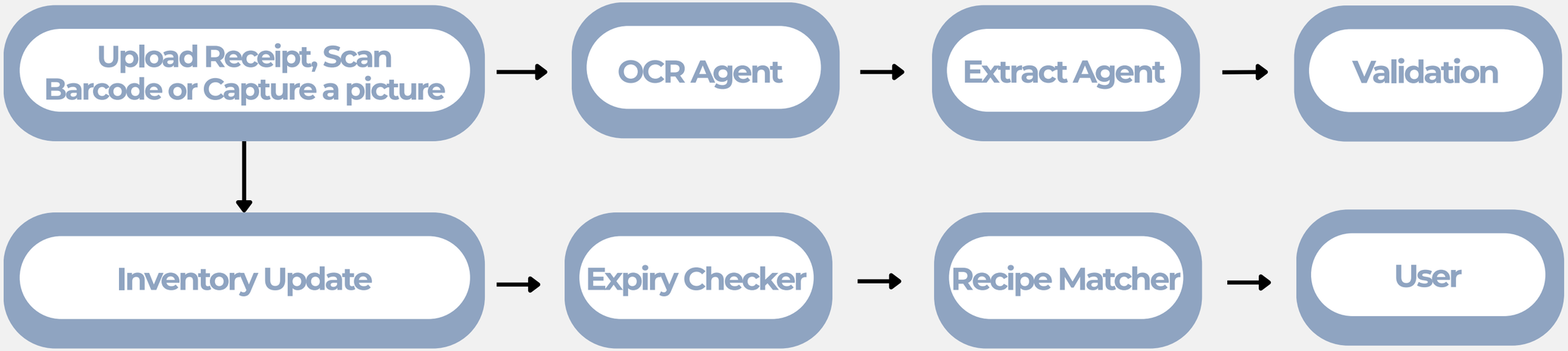
AI Fridge is an **agentic AI system** that automates the entire grocery lifecycle from purchase to consumption. By using **OCR + NLP + multi-agent workflows**, the system extracts items from receipts, tracks inventory states, predicts expiration and recommends recipes using available ingredients. This eliminates manual effort and creates a **closed-loop "purchase-to-pantry-to-plate" ecosystem**.

Main Goal of the System: To automatically track household groceries from purchase to consumption, reducing food waste through intelligent inventory management and recipe suggestions.

Intended Users

- Busy professionals
- Families & households
- Students living independently
- Health-conscious users
- Sustainability-focused consumers

WORKFLOW ORCHESTRATION



PROTOTYPE SUMMARY

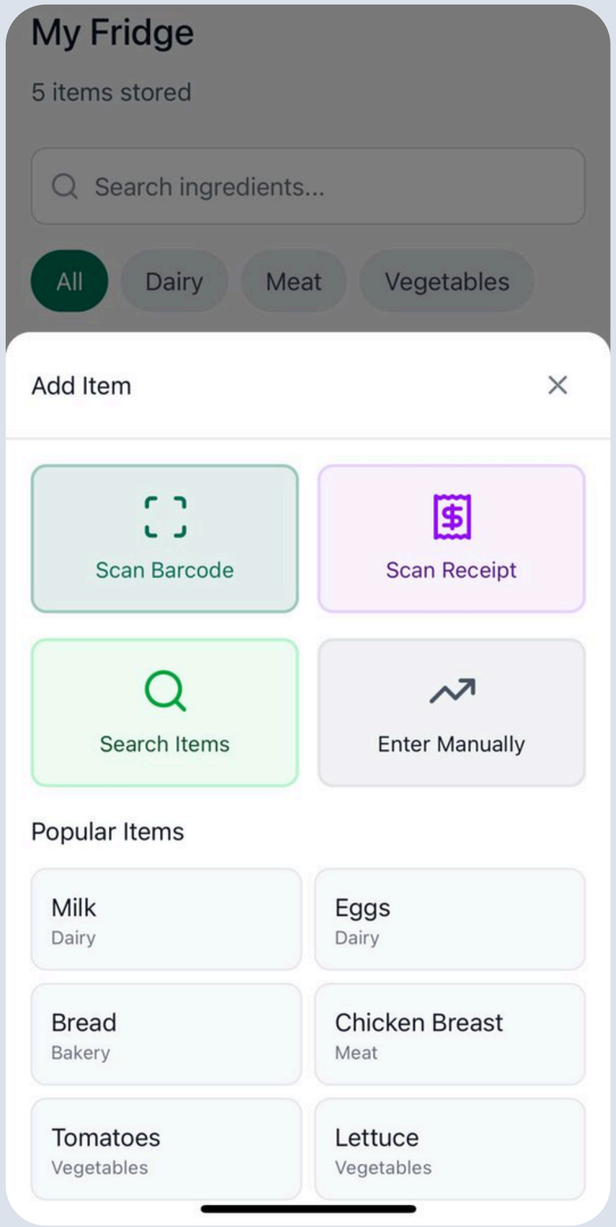
AI Fridge

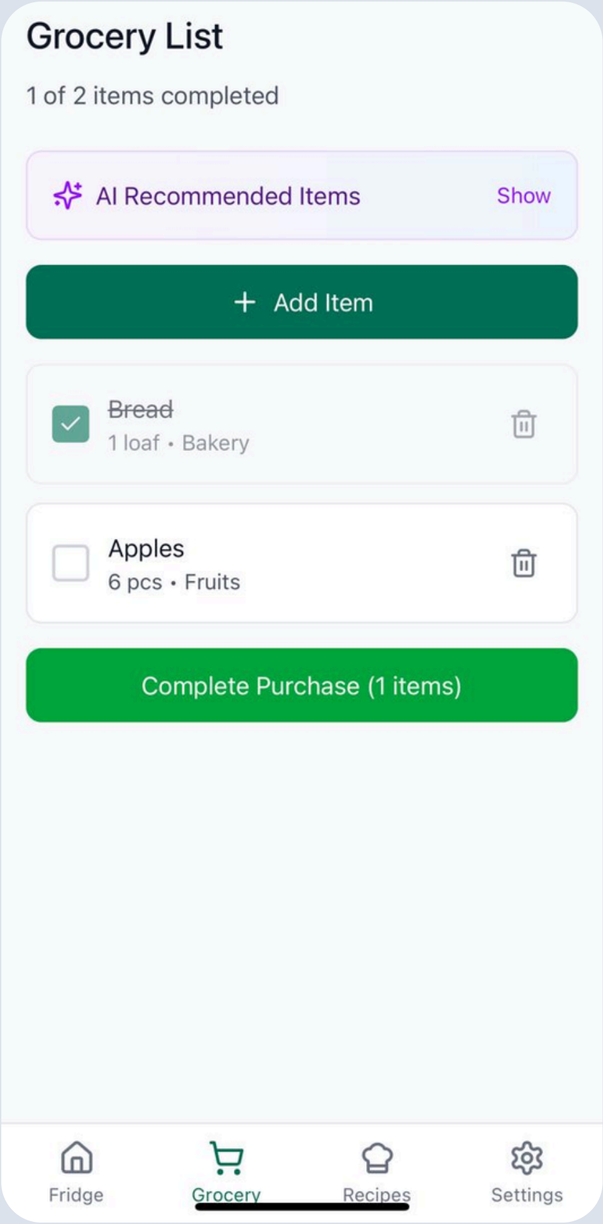
A mobile application that enable users to effectively manage their kitchen ingredients integrated with Z.ai.

Functionality

- **Inventory Management Module:** Manage food items by scanning barcodes, capture a picture of receipts or manually entering item details and track quantity and expiry status integrated with OpenRouter.ai.

- **Shopping List Module:** Manage a shopping list with AI recommendations and automatically transfer purchased items into the inventory after completing a purchase.

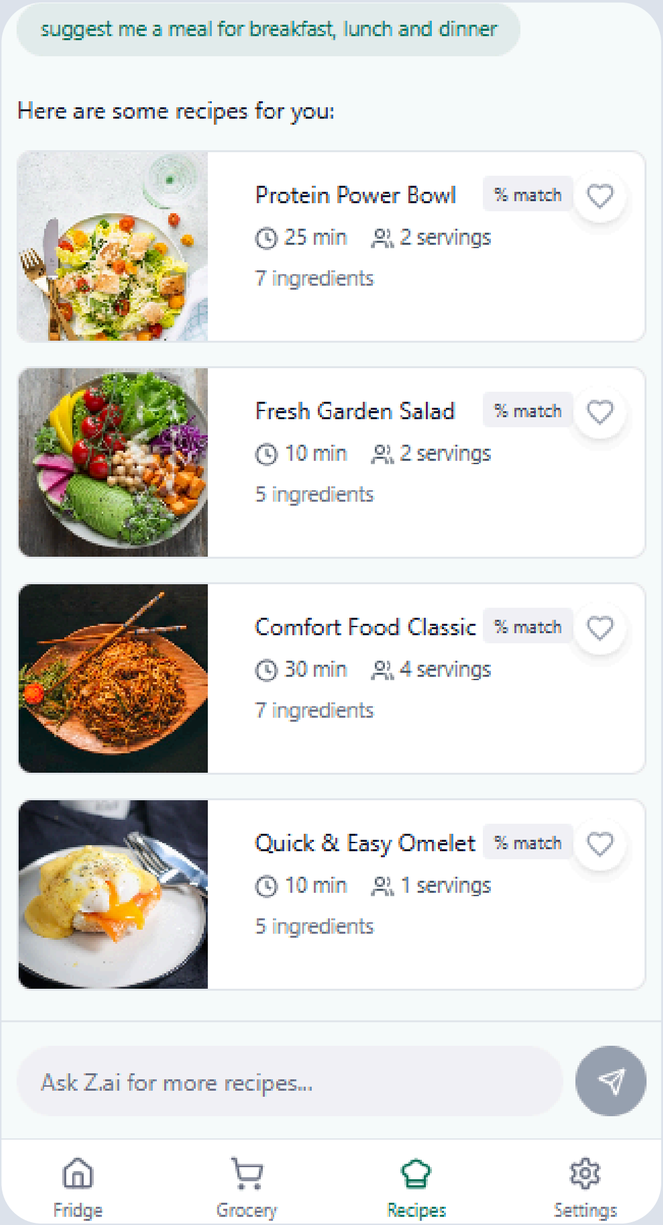


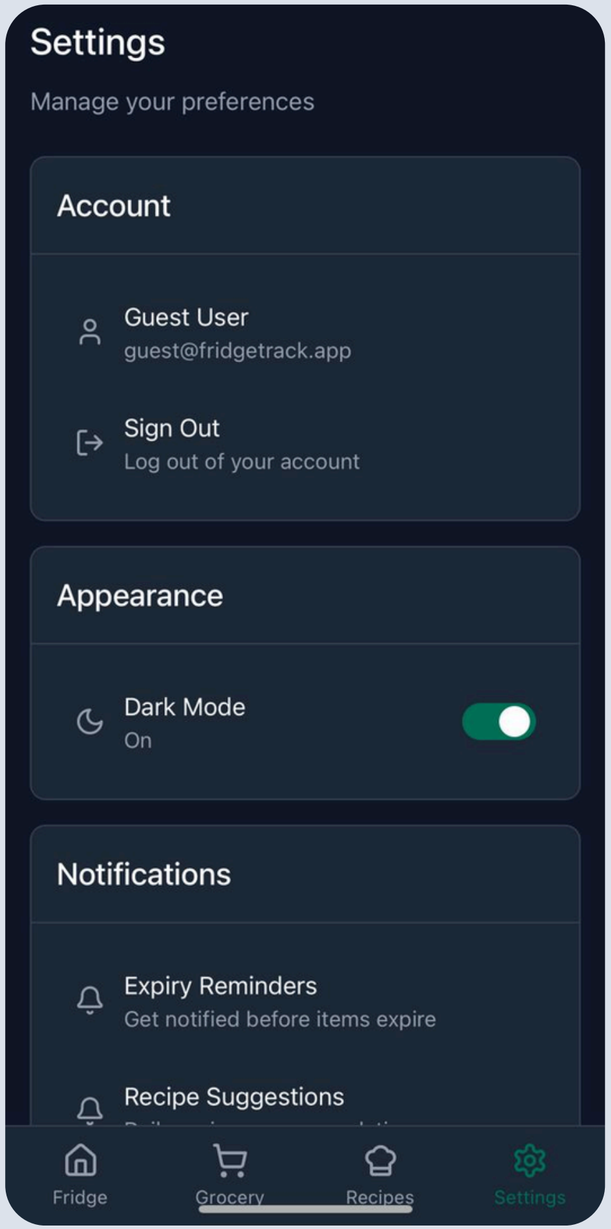


PROTOTYPE SUMMARY

Functionality

- **AI Recipe Recommendation:** An integrated Z.ai chatbot recommends recipes based on current and expiring ingredients that allows interactive user queries and automatically updates inventory after cooking.
- **Dark mode:** Provides a dark mode interface for better usability at night.





Embracing Z.AI

Thank You